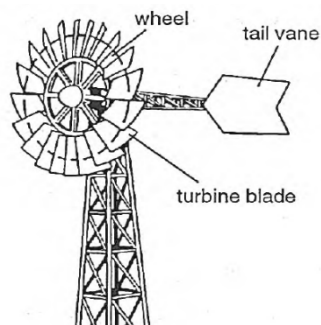
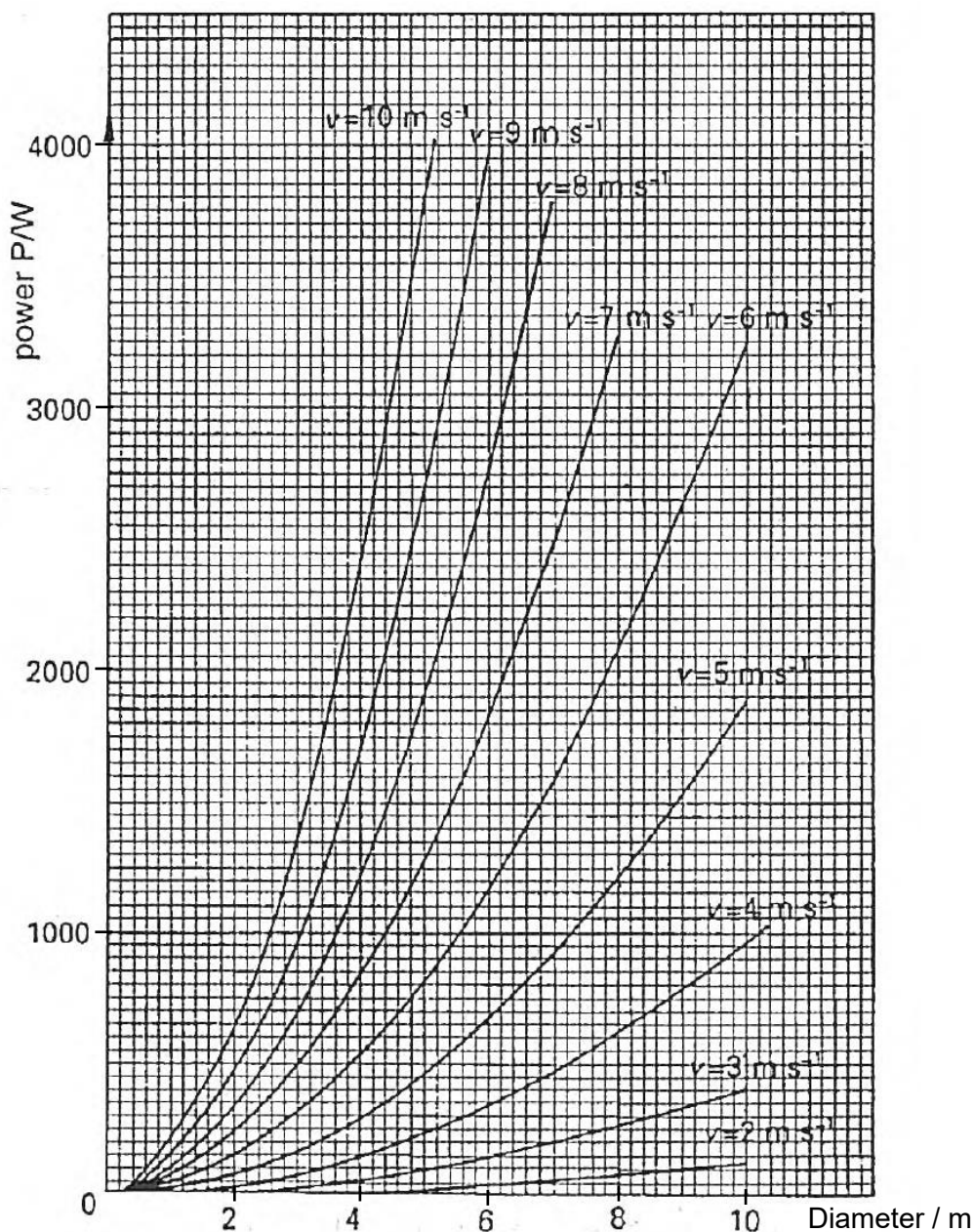


- 8 Wind energy is a renewable source of energy, harnessed from the kinetic energy of moving air. Since the late 1800s, windmills like the one shown in Fig. 8.1 have been used for milling grains.



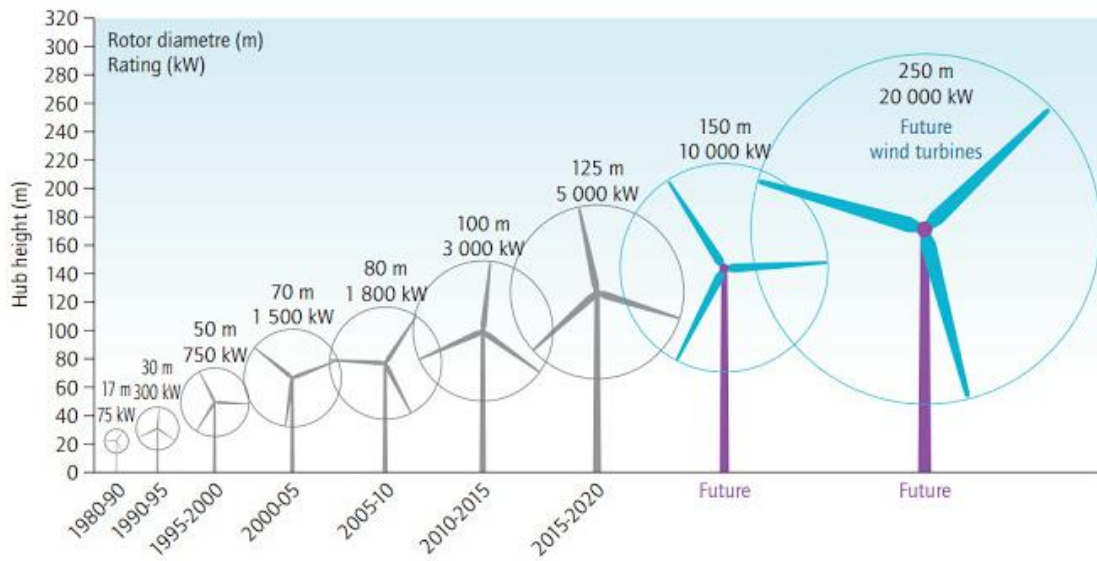
**Fig. 8.1**

Fig. 8.2 shows how the output power of these windmills varies with the overall diameter of the wheel for different wind speeds. The density of air is  $1.3 \text{ kg m}^{-3}$ .



**Fig. 8.2**

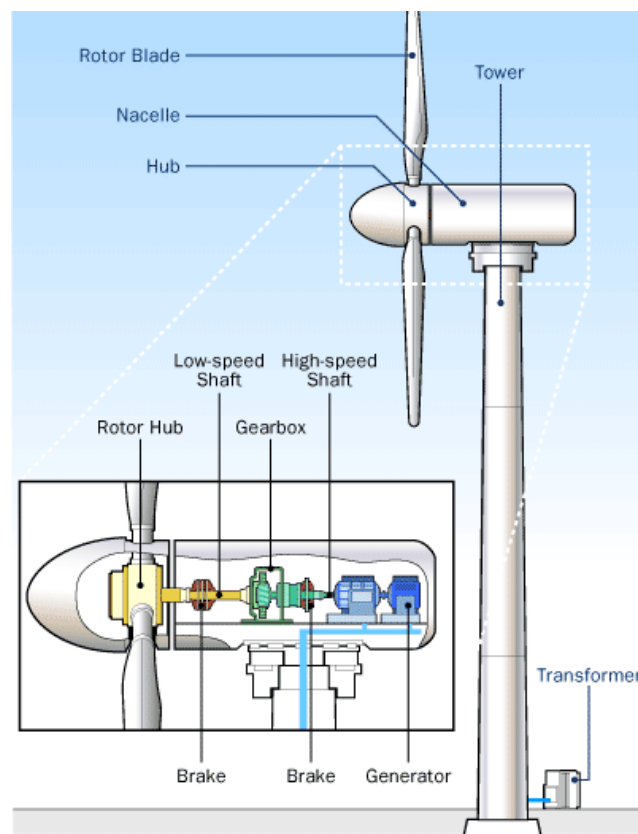
Wind turbines are the modern evolutions of windmills. They have evolved from their multi-bladed predecessors to the modern 3-bladed version. Wind turbines have also increased in hub height and rotor diameter size in the last 45 years as shown in Fig. 8.3.



Source: adapted from EWEA, 2009.

**Fig. 8.3**

Wind turbines have rotor hubs that can change the angle of attack of the rotor blades, which allows it to vary the amount of wind it catches. The nacelle houses a low-speed shaft that is connected to a gearbox which is in turn connected to a high-speed shaft before being connected to a generator. Parts of the wind turbine are shown in Fig. 8.4 below.



**Fig. 8.4**

The choice of location of wind turbines is an important factor to consider when building wind turbines. A table of average wind speed at various heights and locations are shown in Fig. 8.5.

| Height above ground<br>/ m | Wind speed / m s <sup>-1</sup> |          |
|----------------------------|--------------------------------|----------|
|                            | On land                        | Offshore |
| 20                         | 2.43                           | 3.52     |
| 50                         | 3.86                           | 4.51     |
| 75                         | 4.11                           | 5.96     |
| 100                        | 5.29                           | 7.27     |
| 125                        | 6.43                           | 9.51     |

**Fig. 8.5**

- (a) The windmill is most efficient when the wheel and turbine faces the oncoming wind head-on. With reference to Fig. 8.1, explain how the tail vane works.

When the wheel is not facing the wind head-on, a force acts on the tail vane, due to the wind, which provides a moment, turning the wheel to face the wind head-on. [1]

- (b) By considering the kinetic energy possessed by a cylindrical volume of air, prove that the input power  $P$  that can be harnessed by a windmill of cross-sectional area  $A$  is given by

$$P = \frac{1}{2} \rho A v^3,$$

where  $\rho$  is the density of air and  $v$  is the velocity of air. Explain your working.

Consider a cylindrical volume of air of cross-sectional area  $A$ .

The total power  $P$  possessed by this volume of air  $V$  is the total KE of the air in this volume divided by the time it takes for an air particle of velocity  $v$  to travel the length  $L$  of cylinder.

$$P = \frac{E}{t} = \frac{\frac{1}{2}(m)v^2}{t} \quad [1]$$

$$P = \frac{\frac{1}{2}(\rho V)v^2}{t} \quad \text{as } m = \rho V$$

$$P = \frac{\frac{1}{2}(\rho AL)v^2}{t} \quad \text{as the volume of a cylinder is } V = AL \quad [1]$$

$$P = \frac{1}{2} \rho A v^2 \frac{(L)}{t} = \frac{1}{2} \rho A v^3 \quad (\text{shown}) \quad \text{as } v = \frac{(L)}{t} \quad [1]$$

[3]

- (c) (i) For a windmill of diameter 9.0 m, state the power produced by the windmill when the wind speed is  $6.0 \text{ m s}^{-1}$ .

$$\text{Power} = 2650 \text{ W}$$

power = ..... W [1]

- (ii) Calculate the efficiency of a windmill of diameter 9.0 m when the wind speed is  $6.0 \text{ m s}^{-1}$ .

$$\text{Total input power} = P = \frac{1}{2} \rho A v^3 = 0.5(1.3) \pi \left( \frac{9.0}{2} \right)^2 (6)^3 = 8931.9 \text{ W [1]}$$

From Fig. 8.1, a 9.0 m diameter windmill, wind speed of  $6.0 \text{ m s}^{-1}$ ,  $P = 2650 \text{ W}$

$$\text{Efficiency} = \frac{2650}{8931.9} \times 100\% = 29.67 = 30\% [1]$$

efficiency = ..... % [2]

- (iii) Suggest one reason for the loss in efficiency.

Any of the following: 1) Not all energy of air is captured, air still have KE to move past wheel

2) Resistive forces in the gears, 3) there are gaps in the wheel where air flows through

- (d) (i) For the 3-bladed wind turbines built between 2010 to 2015, each typical rotor blade has an effective area of  $120 \text{ m}^2$  that faces the wind head on. The dynamic wind pressure  $p$  is given by

$$p = \frac{1}{2} \rho v^2$$

where  $\rho$  is the density of air and  $v$  is the speed of air.

1. State the hub height of a typical wind turbine built on land between 2010 to 2015.

hub height = 100 m (Fig. 8.3) ..... m [1]

2. By considering the wind speed at the hub height in your answer in (d)(i)1., estimate the moments acting on the wind turbine taken about the base of the tower when built on land.

mtd 1

wind speed at hub height on land =  $5.29 \text{ m s}^{-1}$

Moment =  $F \times \text{hub height } h$

$$= p \times A \times h$$

$$= \frac{1}{2} \times \rho \times v^2 \times A \times h$$

$$= 0.5 \times 1.3 \times 5.29^2 \times (120 \times 3) \times 100$$

$$= 6.54 \times 10^5 \text{ N m}$$

moment = ..... N m [2]

3. Hence suggest why modern wind turbines typically have only 3 blades even though a multi-bladed windmill ensures that more wind energy is harnessed.

Wind blowing on the rotors create a large moment about the base of the turbines. By having fewer blades, it reduces that chance of the wind turbines toppling over / falling / breaking.

[1]

- (e) Explain how an increase in hub height and rotor diameter of wind turbines improves energy production.

Taller towers place the hub height in regions with higher wind speeds (Fig 8.6) [1] Larger

rotors sweep a greater area, which, using  $P = \frac{1}{2} \rho A v^3$ , increases the input power and thus

allows the turbine to produce more energy. [1]

[2]

- (f) (i) Wind turbines typically spin at a rate of 10 to 20 rounds per minute. For a wind turbine with a rotor diameter of 250 m, calculate the speed at the tip of the blade if it were to spin at 30 rounds per minute.

Frequency  $f = 30$  rounds per min  $= 30/60 = 0.50$  Hz [1]

$$v = r\omega = \frac{250}{2} (2\pi f) = 250\pi \times 0.50 = 393 \text{ m s}^{-1} \quad [1]$$

speed = ..... m s<sup>-1</sup> [2]

- (ii) Suggest why this rate of rotation is undesirable.

At 30 rounds per minute, the tip of the blades exceeds the speed of sound. OR This causes structural damage to the blades or produces noise pollution or causes a sonic boom.

[1]

- (g) Explain, using the laws of electromagnetic induction, why there is a need to convert the rate of rotation of the shaft to a high rate before connecting to the generator.

Faraday's law states that the induced emf is proportional to the rate of change of magnetic flux linkage. The gearbox thus allows the generator to spin at higher rates, thus producing a larger emf. [1]

Since  $P = V^2/R$ , this means that more power is generated. [1]

[2]

[Total: 19]

**End of Paper**